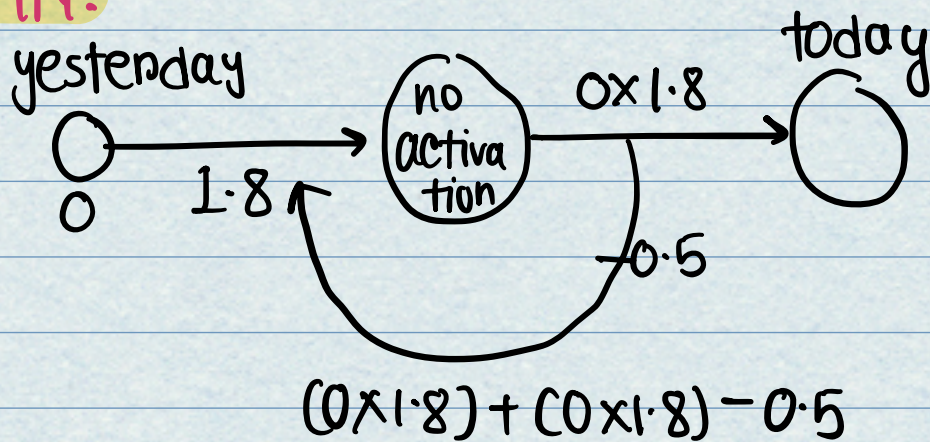
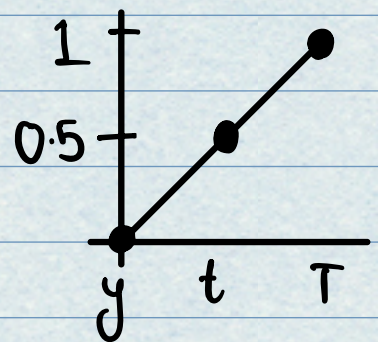
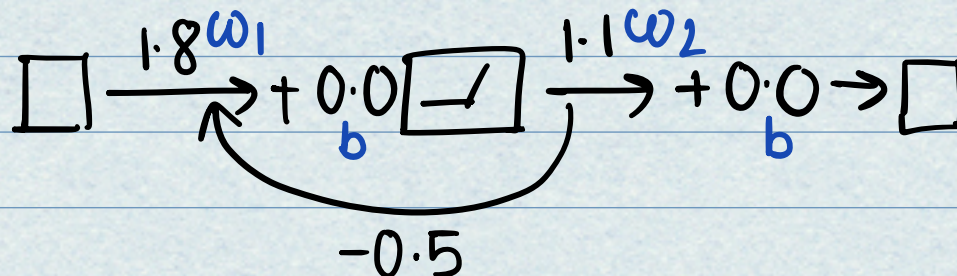


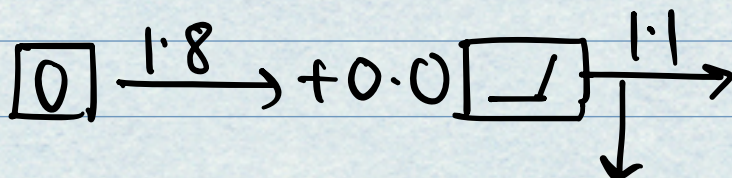
## RNN:



## Stock market Prediction:

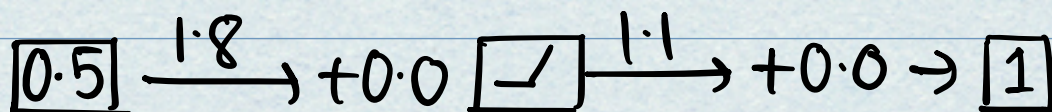


if we unroll this network



$$\text{Relu}(1.8 \times 0 + 0.0) = 0$$

$$0 \times (-0.5) = 0$$

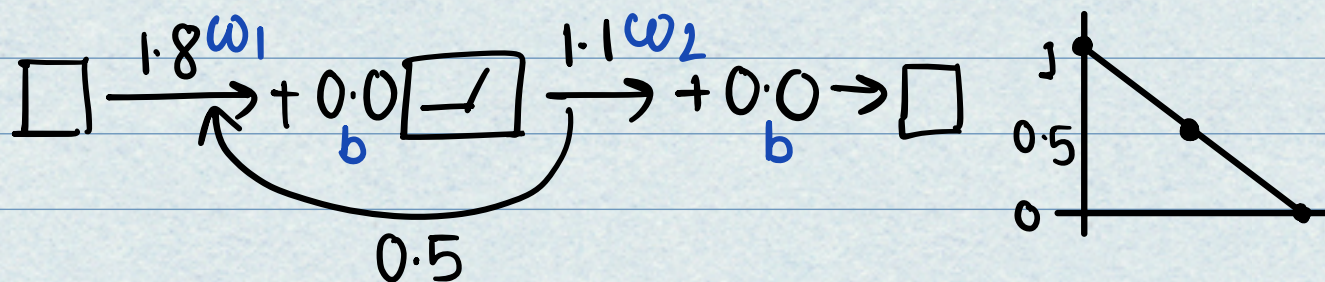


$$(0.5 \times 1.8) = 0.9$$

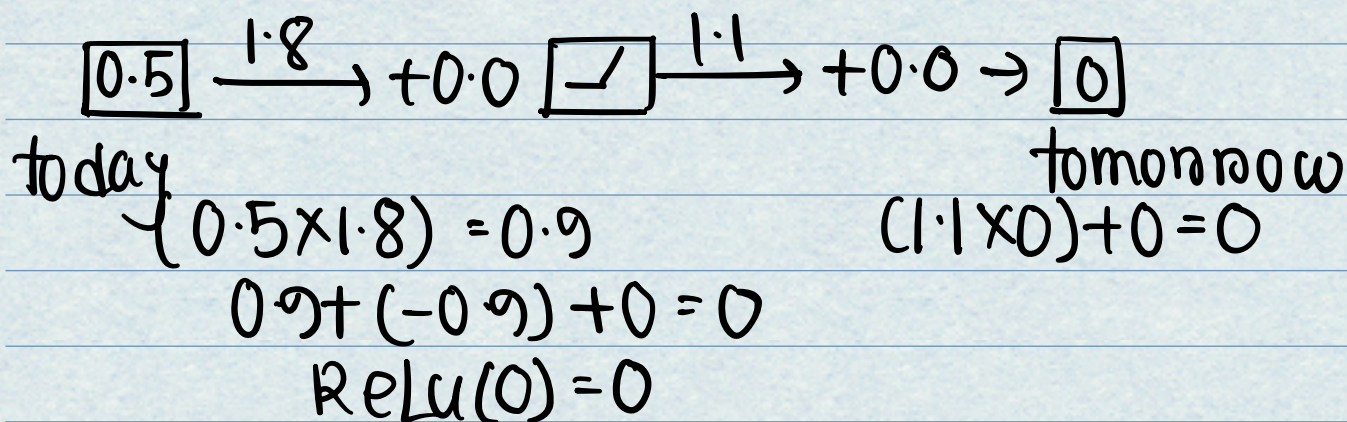
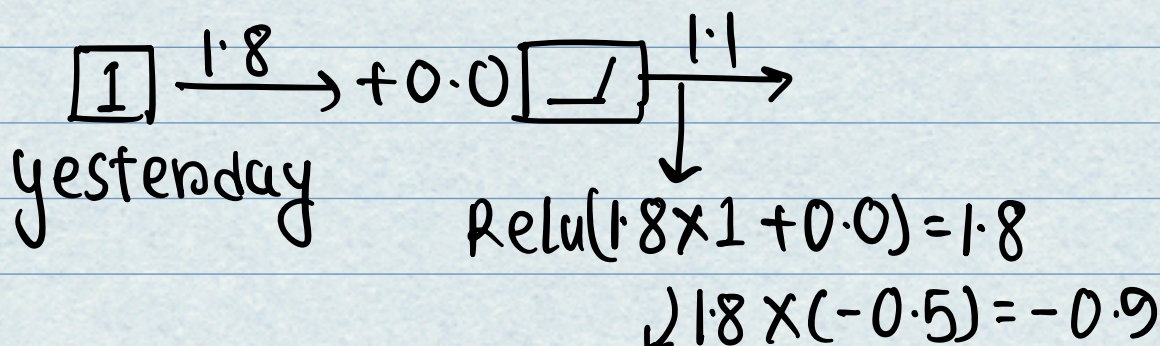
$$0.9 + 0 = 0.9;$$

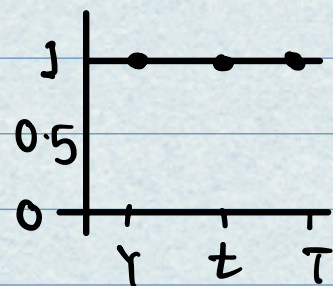
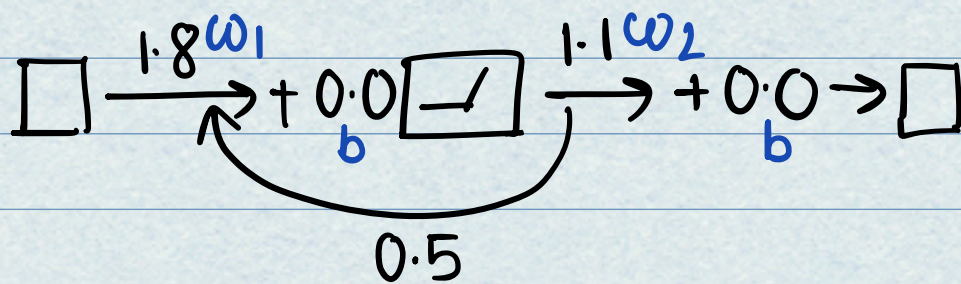
$$\text{Relu}(0.9) = 0.9$$

$$(0.9 \times 1.1) + 0.0 = 0.99 \approx 1$$

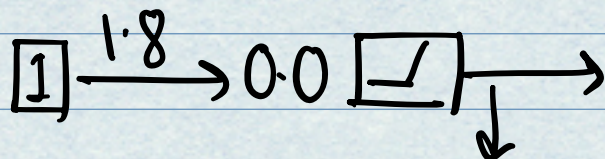


If we unroll this





if we unroll this

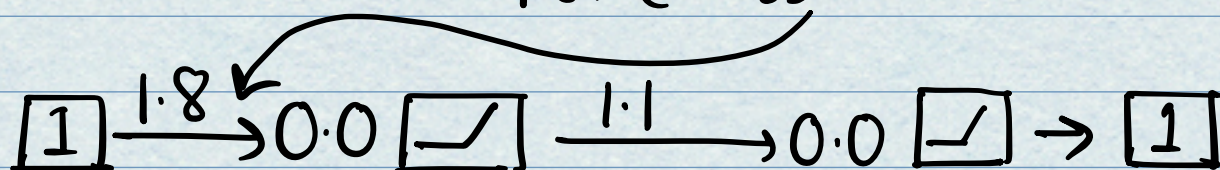


$$(1 \times 1.8) + 0.0 = 1.8$$

$$\text{ReLU}(1.8) = 1.8$$

$$\downarrow$$

$$1.8 \times (-0.5) = -0.9$$



$$(1 \times 1.8) = 1.8$$

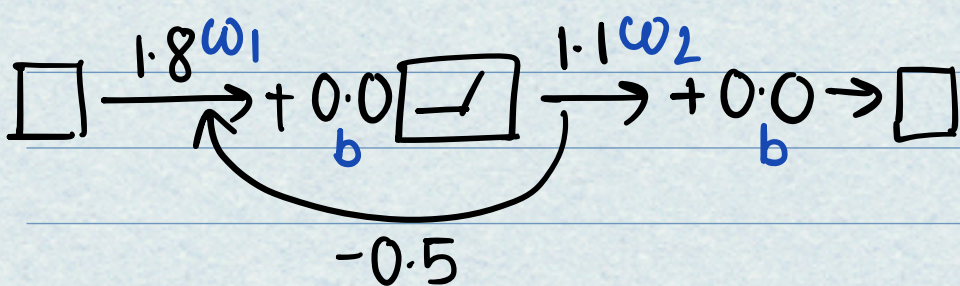
$$-0.9 + 1.8 + 0.0 = 0.9$$

$$\text{ReLU}(0.9) = 0.9$$

$$(0.9 \times 1.1) + 0.0 = 0.99 \approx 1$$

$$\text{ReLU}(1) = 1$$

With Previous 3 day data:



If we unroll this:

$$\boxed{1} \xrightarrow{1.8} + 0.0 \boxed{\phantom{0}} \rightarrow (1 \times 1.8) + 0 = 1.8$$

$$\text{Relu}(1.8) = 1.8$$

$$1.8 \times (-0.5) = -0.9$$

$$\boxed{0.5} \xrightarrow{1.8} + 0.0 \boxed{\phantom{0}} \rightarrow (0.5 \times 1.8) + (-0.9) + 0.0 = 0$$

$$\text{Relu}(0) = 0$$

$$0 \times (-0.5) = 0$$

$$\boxed{0.5} \xrightarrow{1.8} + 0.0 \boxed{\phantom{0}} \xrightarrow{1.1} + 0.0 \boxed{\phantom{0}} \rightarrow \boxed{1}$$

$$(0.5 \times 1.8) + 0 + 0 = 0.9$$

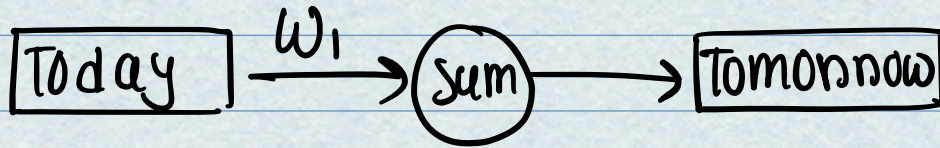
$$\text{Relu}(0.9) = 0.9$$

$$(0.9 \times 1.1) + 0.0$$

$$= 0.99 = 1$$

$$\text{Relu}(1) = 1$$

## Backpropagation: weight and bias update



$$w_1 = w_1 - \eta \frac{\partial \text{loss}}{\partial w_1}$$

$$\text{SSR} = \sum_i^n (\text{Obs} - \text{Predicted})^2$$

$$\frac{\partial \text{loss}}{\partial w_1} = \frac{\partial \text{SSR}}{\partial w_1}$$

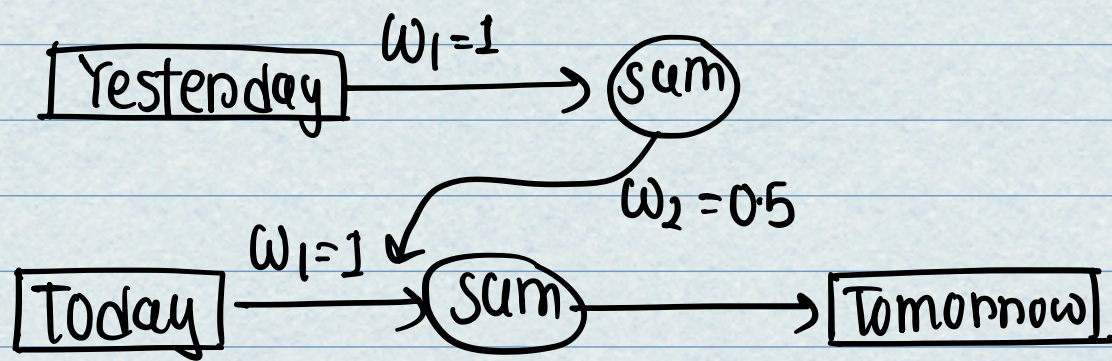
$$\text{predicted} = \text{Today} \times w_1$$

$$= \frac{\partial \text{SSR}}{\partial \text{predicted}} \times \frac{\partial \text{predicted}}{\partial w_1}$$

$$= \frac{\partial}{\partial \text{predicted}} \sum_i^n (\text{Obs}_i - \text{Predicted}_i) \times \frac{\partial}{\partial w_1} (\text{Today} \times w_1)$$

$$= - \sum_i^n 2 (\text{Obs}_i - \text{Predicted}_i) \times \text{Today}$$

$$\therefore \frac{\partial \text{loss}}{\partial w_1} = - \sum_i^n 2 (\text{Obs}_i - \text{Predicted}_i) \times \text{Today}$$



$$\text{Predicted Tomorrow} = \{(\text{Yesterday} \times w_1) \times w_2\} + (\text{Today} \times w_1)$$

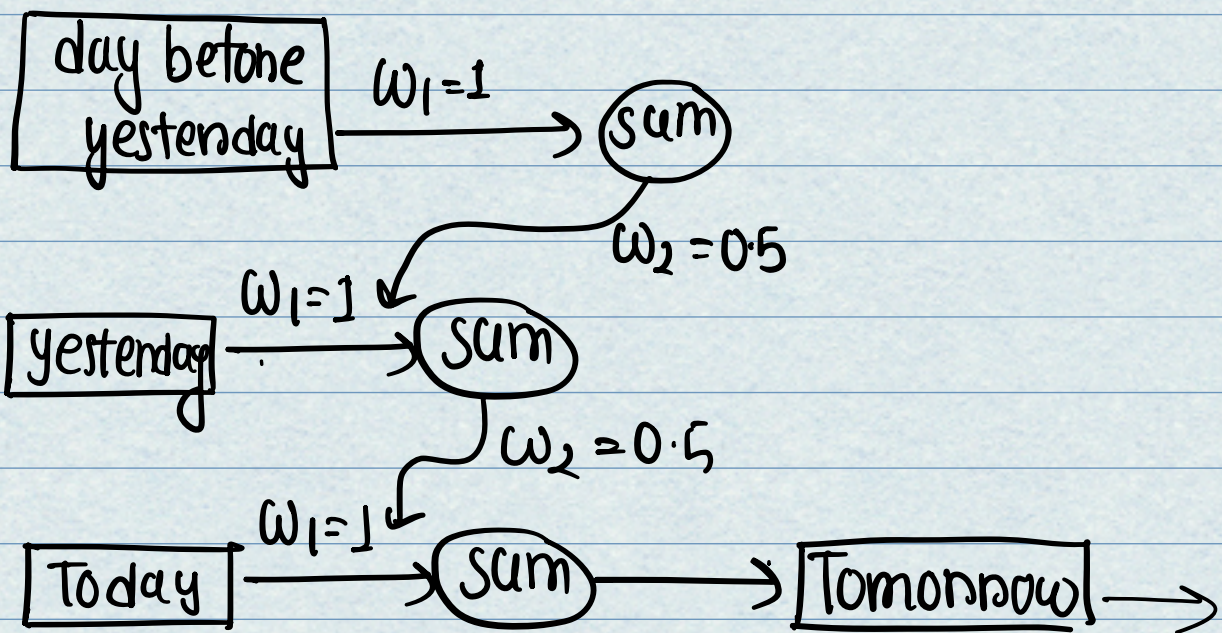
$$\begin{aligned} \frac{\partial \text{loss}}{\partial w_1} &= \frac{\partial \text{SSR}}{\partial w_1} \\ &= \frac{\partial \text{SSR}}{\partial \text{predicted}} \times \frac{\partial \text{predicted}}{\partial w_1} \end{aligned}$$

$$\frac{\partial \text{SSR}}{\partial \text{predicted}} = - \sum_i^n 2 (\text{Obs}_i - \text{Predicted}_i)$$

$$\frac{\partial \text{predicted}}{\partial w_1} = \frac{\partial}{\partial w_1} \{(\text{Yesterday} \times w_1) \times w_2\} + (\text{Today} \times w_1)$$

$$= (\text{Yesterday} \times w_1) + \text{Today}$$

$$\therefore \frac{\partial \text{loss}}{\partial w_1} = - \sum_i^n 2 (\text{Obs}_i - \text{Predicted}_i) \times \{(\text{Yesterday} \times w_1) + \text{Today}\}$$



$$\frac{\partial \text{SSR}}{\partial \text{predicted}} = - \sum_i^n (\text{Obs}_i - \text{Predicted}_i)$$

$$\text{Predicted} = [\{(\text{D.B.Y} \times w_1)w_2\} + (\text{Yesterday} \times w_1)]w_2 + (\text{Today} \times w_1)$$

$$\frac{\partial \text{predicted}}{\partial w_1} = \frac{\partial}{\partial w_1} [\{(\text{D.B.Y} \times w_1)w_2\} + (\text{Yesterday} \times w_1)]w_2 + (\text{Today} \times w_1)$$

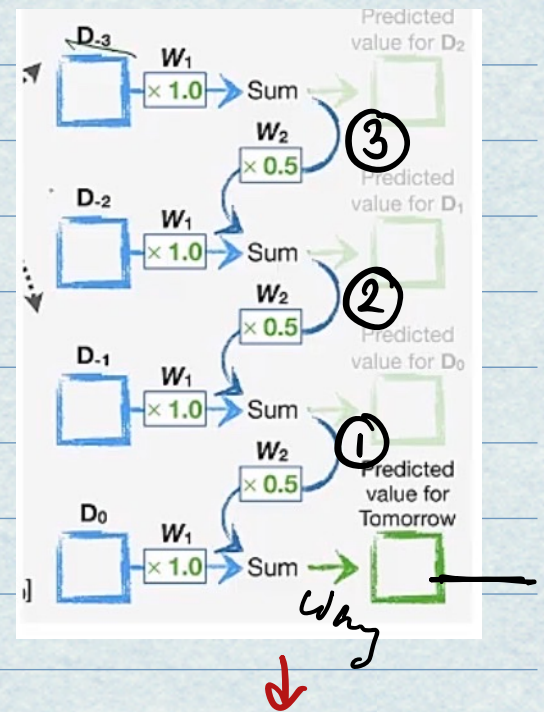
$$= [(\text{D.B.Y} \times w_2) + \text{Yesterday}]w_2 + \text{Today}$$

$$= (\text{D.B.Y} \times w_2^2) + (\text{Yesterday} \times w_2) + \text{Today}$$

$$\therefore \frac{\partial \text{loss}}{\partial w_1} = - \sum_i^n (\text{Obs}_i - \text{Predicted}_i) \times (D \cdot B \cdot Y \times w_2^2) + (\text{Yesterday} \times w_2) + \text{Today}$$

This is called backpropagation by time

$$\frac{\partial \text{predicted}}{\partial w_1} = (D_{-3} \times w_2^3) + (D_{-2} \times w_2^2) + (D_{-1} \times w_2) + D_0$$

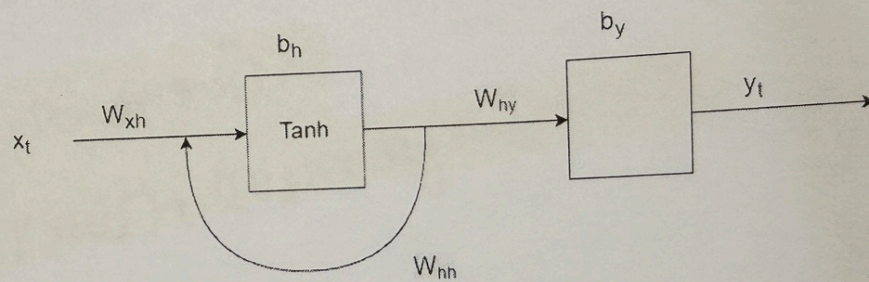


Vanishing gradient problem

Exploding gradient problem

ଅବଦାନ unfold  
ତେଜ power?

Question 1:



$W_{xh} = 0.4 \mid W_{hh} = 0.3 \mid W_{hy} = 4.0 \mid b_h = 0.1 \mid b_y = 1.5$   
 Temperature readings for four days are  $8^\circ\text{C}, 5^\circ\text{C}, 2^\circ\text{C}, 4^\circ\text{C}$ .

$$\text{Tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- Predict the temperature of the 5th day.
- Assume, the recurrent unit is not using any activation function, find the expression for  $\frac{d\text{Loss}}{dW_{xh}}$ .
- If  $|W_{hh}| > 1$  What issue will this network face?

$$\begin{aligned} \text{a)} \quad (8 \times 0.4) + 0.1 &= 3.3 ; \tanh(3.3) = 0.997 \\ &= 0.997 \times 0.3 \\ &= 0.2991 \end{aligned}$$

$$(5 \times 0.4) + 0.2991 + 0.1 = 2.3991 ;$$

$$\begin{aligned} \tanh(2.3991) &= 0.9836 \\ 0.9836 \times 0.3 &= 0.295 \end{aligned}$$

$$(2 \times 0.4) + 0.295 + 0.1 = 1.195 ;$$

$$\begin{aligned} \tanh(1.195) &= 0.83 \\ 0.83 \times 0.3 &= 0.249 \end{aligned}$$

$$(4 \times 0.4) + 0.249 + 0.1 = 1.94 ;$$

$$\tanh(1.94) = 0.9604$$

$$(0.9604 \times 4) + 1.5 = 5.3416$$

Shortcut way:

$$\tanh((0.4 \times 8) + (0.3 \times 0) + 0.1) = \tanh(3.3) = 0.996$$

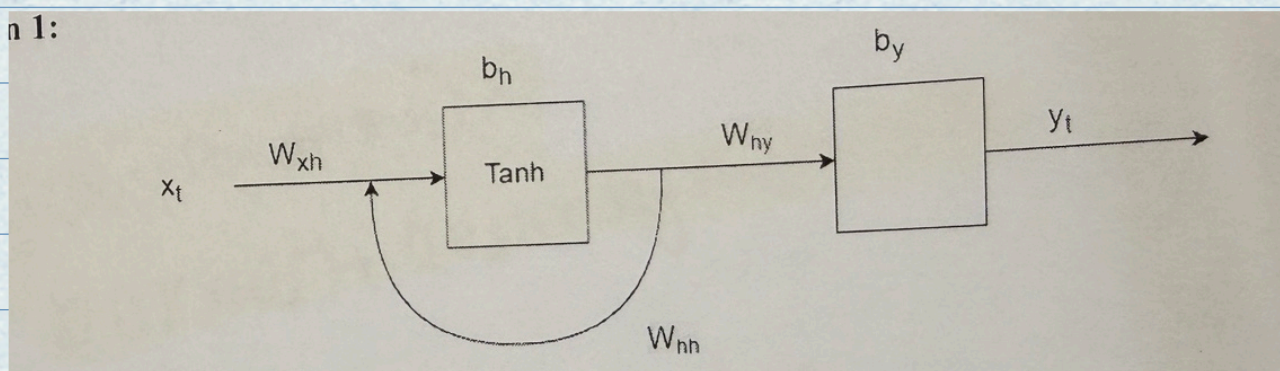
$$\tanh((0.4 \times 5) + (0.3 \times 0.996) + 0.1) = \tanh(2.39) = 0.98$$

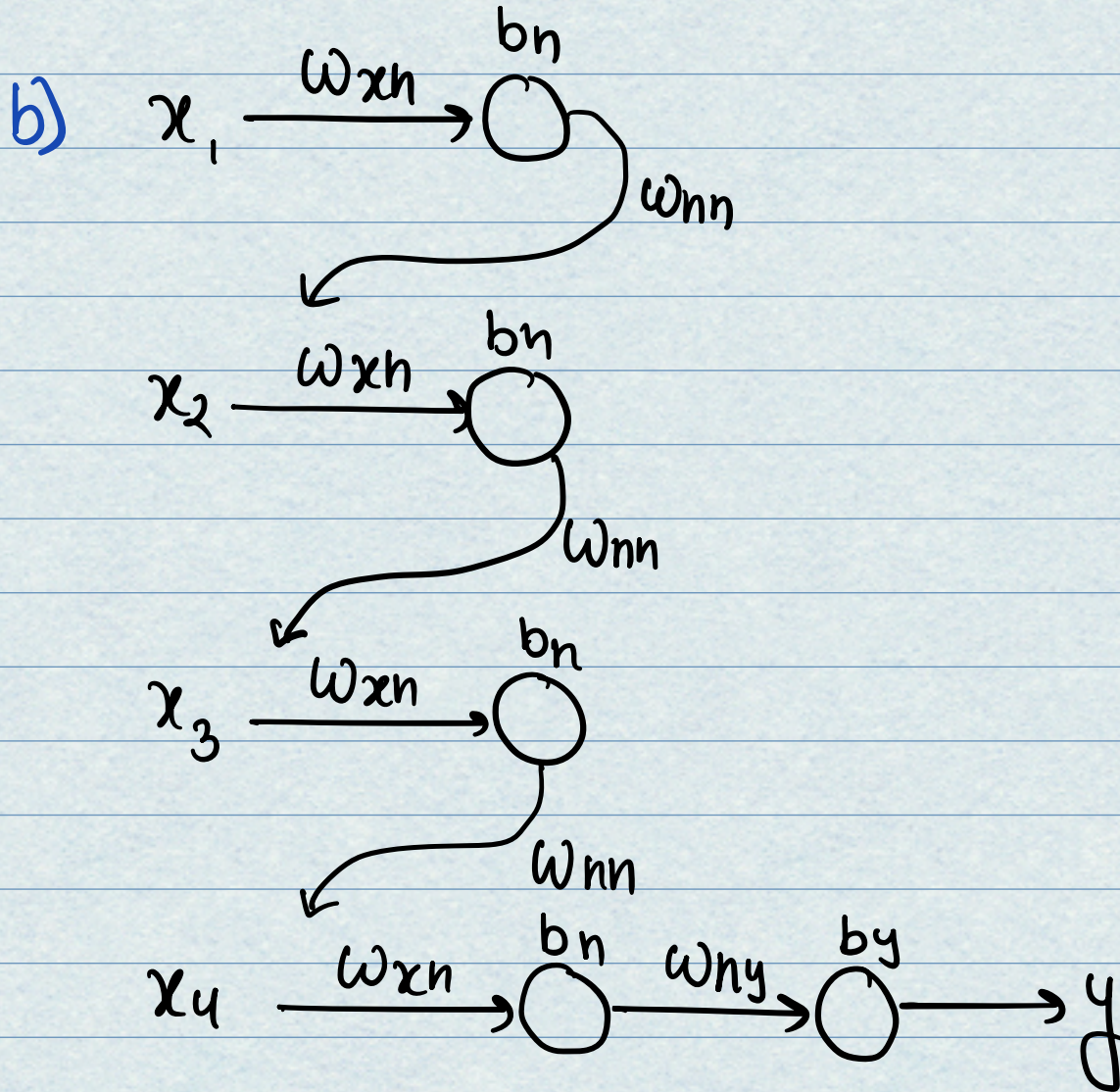
$$\tanh((0.4 \times 2) + (0.3 \times 0.98) + 0.1) = \tanh(1.195) = 0.83$$

$$\tanh((0.4 \times 4) + (0.3 \times 0.83) + 0.1) = \tanh(1.94) = 0.96$$

$$(0.9604 \times 4) + 1.5 = 5.3416$$

n 1:





$$\text{Predicted} = \left\{ \left[ \underbrace{(\omega_{xn} x_1 + b_n)}_{\text{blue}} \omega_{nn} + \underbrace{(\omega_{xn} x_2 + b_n)}_{\text{green}} \right] \omega_{nn} + \underbrace{(\omega_{xn} x_3 + b_n)}_{\text{pink}} \omega_{nn} + \underbrace{(\omega_{xn} x_4 + b_n)}_{\text{purple}} \right\} \omega_{ny} + b_y$$

$$\begin{aligned} \frac{\partial \text{Predicted}}{\partial \omega_{xn}} &= \left[ \left[ \underbrace{x_1 \omega_{nn}}_{\text{blue}} + \underbrace{x_2}_{\text{green}} \right] \omega_{nn} + \underbrace{x_3}_{\text{pink}} \omega_{nn} + \underbrace{x_4}_{\text{purple}} \right] \omega_{ny} \\ &= \left[ x_1 \omega_{nn}^2 + x_2 \omega_{nn} + x_3 \omega_{nn} + x_4 \right] \omega_{ny} \\ &= \left[ x_1 \omega_{nn}^3 + x_2 \omega_{nn}^2 + x_3 \omega_{nn} + x_4 \right] \omega_{ny} \end{aligned}$$

$$\frac{\partial \text{loss}}{\partial w_{xn}} = \frac{\partial \text{loss}}{\partial \text{predicted}} \times \frac{\partial \text{predicted}}{\partial w_{xn}}$$

$$\frac{\partial \text{loss}}{\partial \text{predicted}} = \frac{d}{dPn} \sum (Ob - \text{Predicted})^2$$

$$= -2 \sum (Ob - \text{predicted})$$

$$\therefore \frac{\partial \text{loss}}{\partial w_{xn}} = \{-2 \sum (Ob - \text{predicted})\} \times$$

$$[x_1 w_{nn}^3 + x_2 w_{nn}^2 + x_3 w_{nn} + x_4] w_{ny}$$